30/05/2026

Code-P

Aakash
Medical | IIT-JEE | Foundations

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Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **Project Phoenix for Re-NEET-2026_CST-03P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. The displacement of a moving particle is given by, $x = at^3 + bt^2 + ct + d$. The acceleration of particle at $t = 3$ s would be

(1) $2(a + 9b)$

(2) $2(9a + b)$

(3) $9(a + b)$

(4) $3(2a + b)$

2. A particle moves with velocity $\vec{v}$ in xy plane as $\vec{v} = c\left(xy^2\hat{i} + yx^2\hat{j}\right)$ m/s . Then the general equation of its path will be

(1) $y = x$ + constant

(2) $y = x^2$ + constant

(3) $y^2 = x^2$ + constant

(4) $y^3 = x^3$ + constant

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3. Two balls are thrown vertically up simultaneously with different initial speeds. The relative velocity of one ball w.r.t. the other as long as they are in air

(1) Decreases with time

(2) Increases with time

(3) First increases then decreases with time

(4) Remains constant

4. A body is projected with velocity $\vec{v} = \left(\alpha\hat{i} + \beta\hat{j}\right)$ m/s. The time of flight of body is [considering x as horizontal and y as vertical axis and g is acceleration due to gravity]

(1) $\frac{2\beta}{g}$

(2) $\frac{2\alpha}{g}$

(3) $\frac{2\alpha\beta}{g}$

(4) $\frac{2\alpha}{g\beta}$

5. In an experiment refractive index of glass was observed to be 1.45, 1.56, 1.54, 1.44, 1.54 and 1.53. The mean absolute error in the experiments is

(1) 0.04

(2) 0.02

(3) 0.03

(4) 0.01

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6. A body of mass $\frac{1}{\sqrt{2}}$ kg moving along *x*-axis at 4 m/s, suddenly takes a right turn and starts moving along *y*-axis at the same speed. The magnitude of change in momentum of the body is

(1) 5 kg m/s

(2) $8\sqrt{2}$ kg m/s

(3) 4 kg m/s

(4) $2\sqrt{2}$ kg m/s

7. The upper half of an inclined plane of inclination 'θ' is frictionless while lower half is rough. A body start to slip from top of inclined surface and come to rest at bottom. The coefficient of friction (μ) at lower half of plane is

(1) 2 cotθ

(2) cotθ

(3) tanθ

(4) 2 tanθ

8. A force $\vec{F} = \left(2x\hat{i} + 3\hat{j} + 3z^2\hat{k}\right)$ is acting on the particle of mass 2 kg which is displaces from point A (2, 3, 0) to point B (3, 2, 1). The work done by the force on the particle will be

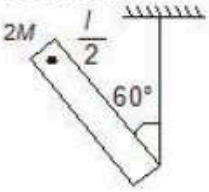
(1) 6 J

(2) 3 J

(3) 4 J

(4) 2 J

9. If a rod of mass 2M and length $\frac{l}{2}$ is hinged at one end and the other end is hanged with the help of the string as shown in the figure.
If string is cut at $t = 0$. Find the angular velocity when the rod becomes vertical.



(1) $\sqrt{\frac{3g}{l}}$

(2) $\sqrt{\frac{2g}{l}}$

(3) $\sqrt{\frac{3g}{2l}}$

(4) $\sqrt{\frac{2g}{3l}}$

10. A : The working of venturimeter is based on Bernoulli's theorem.
R : Bernoulli's theorem states that sum of kinetic energy per unit volume and potential energy per unit volume of a fixed a mass of liquid is constant

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

11. The escape velocity from the earth surface is about 11.2 km/s. The escape velocity from surface of a planet having twice the radius and the same mean density as that of earth, will be

(1) 11.2 km/s

(2) 22.4 km/s

(3) 5.6 km/h

(4) 18.7 km/h

12. Consider the following statements:
Statement (A): Bulk modulus is relevant for solids, liquids and gases.
Statement (B) : Shear modulus is only relevant for liquids.
Statement (C): Young's modulus is relevant for solids only.
The **correct** statement(s) is/are

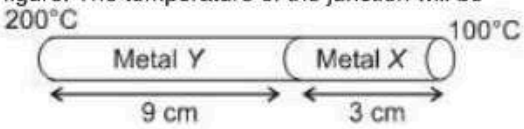
(1) Only (A)

(2) Only (A) and (B)

(3) Only (C)

(4) Only (A) and (C)

13. The coefficient of thermal conductivity of metal *X* is 3 times that of metal *Y* in the composite cylindrical bar shown in the figure. The temperature of the junction will be



(1) 90°C

(2) 110°C

(3) 150°C

(4) 135°C

14. The temperature of ideal gas is increased from –73°C to 127°C, then approximate percentage increase in its average speed is

(1) 41%

(2) 50%

(3) 82%

(4) 100%

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15. A Carnot engine takes heat from a reservoir at 327°C and gives out heat to the sink at 27°C. The efficiency of the engine is

(1) 50%

(2) 60%

(3) 70%

(4) 55%

16. Starting from same initial conditions an ideal gas expands from volume V_1 to V_2 in three different ways. The work done is W_1 in isothermal process, W_2 in adiabatic process and W_3 in isobaric process. Then

(1) $W_1 > W_2 > W_3$

(2) $W_3 > W_2 > W_1$

(3) $W_3 > W_1 > W_2$

(4) $W_2 > W_3 > W_1$

17. The wave function of a pulse is given by $y = \frac{x}{x^2+1}$, where x and y are in meter. The propagation velocity of the wave is 2 m/s in negative x-direction, then the equation of wave at any time t is given as

(1) $y = \frac{x+2t}{(x+2t)^2+1}$

(2) $y = \frac{x+\frac{t}{2}}{\left(x+\frac{t}{2}\right)^2+1}$

(3) $y = \frac{\left(x-\frac{t}{2}\right)}{\left(x-\frac{t}{2}\right)^2+1}$

(4) $y = \frac{x-2t}{(x-2t)^2+1}$

18. The equation of simple harmonic motion is given by $y = 20\sin\left[\frac{4\pi t}{15} + \theta\right]$ (Where y is in cm and t is in second). If the displacement at time t = 0, is $\frac{20}{\sqrt{2}}$ cm, then the total phase at t = 1.5 s

(1) $\frac{\pi}{20}$ rad

(2) $\frac{3}{20}\pi$ rad

(3) $\frac{13}{10}\pi$ rad

(4) $\frac{13}{20}\pi$ rad

19. A particle of mass 100 g is executing SHM of amplitude 10 cm. When the particle passes through mean position, its kinetic energy is 8 mJ. The equation of motion of particle if its initial phase is $\frac{\pi}{4}$ rad is

(1) $y = 0.1\sin\left(2t + \frac{\pi}{4}\right)$ m

(2) $y = 0.1\sin\left(t + \frac{\pi}{4}\right)$ m

(3) $y = 0.1\sin(2\pi + 2t)$ m

(4) $y = 0.1\sin\left(4t + \frac{\pi}{4}\right)$ m

20. In a resonance tube experiment, the first two resonance are observed at length 10.5 cm and 29.5 cm. The third resonance is observed at the length

(1) 47.5 cm

(2) 58.5 cm

(3) 48.5 cm

(4) 82.8 cm

21. A spring of force constant 800 N/m has an extension of 5 cm. The work done by the external Agent in stretching the spring from 5 cm to 15 cm is

(1) 16 J

(2) 8 J

(3) 32 J

(4) 24 J

22. A cylinder of height 20 m is completely filled with water. The velocity of efflux of water through a hole on the side wall at a depth of 5 m from free surface is (g = 10 m/s^2)

(1) 10 m/s

(2) 20 m/s

(3) 15 m/s

(4) 17.3 m/s

23. Consider following statements out of which one is labelled as assertion and other as reason.
Assertion (A): Electrostatic field lines are normal to a conducting surface.
Reason (R): A conducting surface is not an equipotential surface in electrostatic equilibrium.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

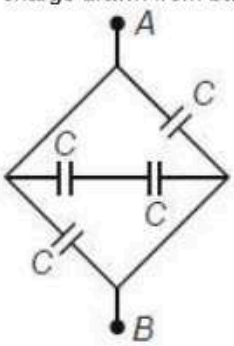
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

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24. Four identical capacitors are connected as shown in figure. When a battery of 6 V is connected between A and B, the charge drawn from battery is 1.5 μC. The value of C is



(1) 2.5 μF

(2) 1 μF

(3) 1.5 μF

(4) 0.1 μF

25. The number of excess electrons present in a body having net charge -1.6×10^{-18} C is

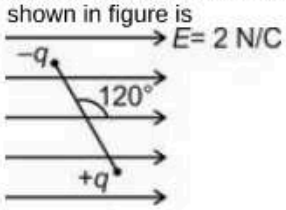
(1) 100

(2) 5

(3) 1

(4) 10

26. The torque acting on a dipole of dipole moment 10^{-8} C-m placed inside a uniform electric field E = 2 N/C at an instant shown in figure is



(1) 10^{-8} N-m

(2) 2×10^{-8} N-m

(3) $\sqrt{3} \times 10^{-6} \mathrm{N-m}$

(4) $\sqrt{3} \times 10^{-8} \mathrm{N-m}$

27. A charged particle is at rest in a space, where electromagnetic field is present. It experiences electromagnetic force, then

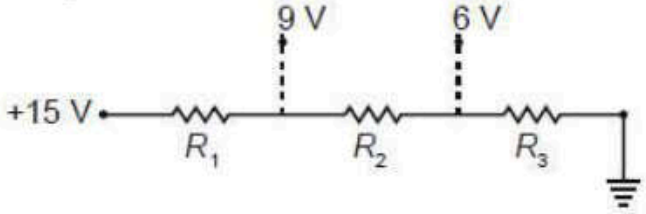
(1) The electric field must be zero

(2) Electric field must be non-zero

(3) The electric field may be zero

(4) The magnetic field must be zero

28. A potential divider is used to give outputs of 6 V and 9 V from a 15 V source as shown in figure. The value of R_1, R_2 and R_3 may be respectively



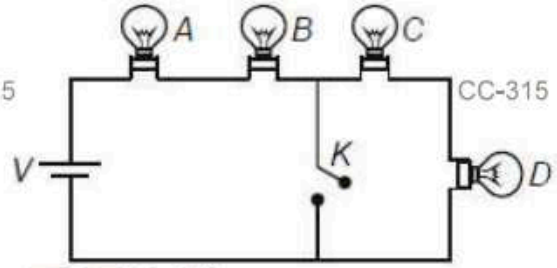
(1) 1 kΩ, 2 kΩ and 3 kΩ

(2) 100 Ω, 100 Ω and 300 Ω

(3) 200 Ω, 100 Ω and 200 Ω

(4) 10 Ω, 20 Ω and 30 Ω

29. Four identical bulbs each of rated power P and voltage V are connected as shown in figure. Now after the key K is closed then incandescence of bulb A will



(1) Increase

(2) Decrease

(3) Become zero

(4) Remain same

30. Match the column-I with column-II

Column-I	Column-II
a. Permeability of free space	(i) $[MT^{-2}A^{-1}]$
b. Magnetic field	(ii) $[L^2A]$
c. Magnetic moment	(iii) $[ML^2T^{-2}]$
d. Torsion constant	(iv) $[MLT^{-2}A^{-2}]$

(1) a(i), b(ii), c(iv), d(iii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(iv), b(i), c(iii), d(ii)

31. By changing current from 5 A to 10 A in 0.1 s the self induced emf is 10 V. The change of energy of the magnetic field of the coil will be

(1) 5 J

(2) 6 J

(3) 7.5 J

(4) 9 J

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32. In an alternating current circuit consisting of some elements in series, the current increases on increasing the frequency of the supply. Which of the following elements are likely to constitute the circuit?

(1) Only resistor

(2) Resistor and capacitor

(3) Only inductor

(4) Resistor and inductor

33. An a.c. source is $100\sqrt{2}$ V, 50 Hz. The value of voltage after $\frac{1}{600}$ second after zero value is

(1) 50 V

(2) 100 V

(3) 150 V

(4) 200 V

34. In YDSE, a thin glass slab (μ = 1.5) is introduced in the path of one of the two interfering beams. The central fringe moves to a position occupied by the fifth bright fringe earlier. Thickness of the glass slab is (λ = 5000 Å)

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(1) 4000 Å

(2) 6000 Å

(3) 2000 Å

(4) 50000 Å

35. An unpolarised beam of light is incident on a set of four polaroids such that axis of each polaroids makes an angle of $\frac{\pi}{3}$ with preceding sheet. The ratio of intensity of light transmitted to the intensity of light incident is

(1) $\frac{1}{32}$

(2) $\frac{1}{64}$

(3) $\frac{1}{256}$

(4) $\frac{1}{128}$

36. An air bubble inside a glass slab (μ = 1.4) appears 4 cm when viewed from one side and 6 cm from when viewed from other side (opposite). The thickness of the slab is

(1) 15 cm

(2) 14 cm

(3) 10 cm

(4) 12 cm

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37. A glass equi-convex lens of refractive index (3/2) has focal length *f* in air. If the lens is immersed in water of refractive index (4/3) its new focal length would be

(1) 1.2*f*

(2) 2*f*

(3) 4*f*

(4) 2.5*f*

38. The refractive indices of violet and red light are 1.54 and 1.52 respectively. If the angle of thin prism is 6°, then angular dispersion is

(1) 0.2°

(2) 0.12°

(3) 0.36°

(4) 0.6°

39. The relative permeability of a liquid is $\frac{3}{16}$ and relative permittivity is 16. Then the refractive index of the liquid is

(1) 1.41

(2) 1.73

(3) 0.11

(4) 3

40. The work function of a photosensitive material is 4.2 eV. The wavelength of incident radiation on the cathode of this material, for which the stopping potential is 2.0 V lie in the

(1) X–ray region

(2) Ultraviolet region

(3) Visible region

(4) Infrared region

41. According to Bohr's theory, maximum frequency of emission is obtained for the transition

(1) $n = 2$ to $n = 1$

(2) $n = 6$ to $n = 2$

(3) $n = 3$ to $n = 2$

(4) $n = 6$ to $n = 5$

42. The ${}_{92}U^{238}$ nucleus, hypothetically undergoes successively 9α and $8\beta^-$ decays. The resulting nucleus will be

(1) ${}_{82}Pb^{202}$

(2) ${}_{82}Pb^{220}$

(3) ${}_{94}Pu^{238}$

(4) ${}_{93}Np^{232}$

43. The p-type semiconductor is

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(1) Negatively charged

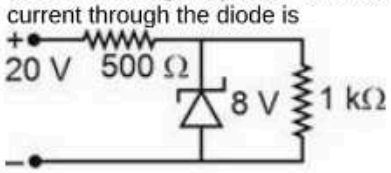
(2) Positively charged

(3) Electrically neutral

(4) May be negatively charged or positively charge

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44. A zener diode having breakdown voltage equal to 8 V is used in a voltage regulator circuit as shown in the figure. The current through the diode is



(1) 8 mA

(2) 16 mA

(3) 24 mA

(4) 12 mA

45. Read the statements A and B carefully and select the correct option.

Statement A: In the absence of an applied voltage, the net flow of charge in any one direction for a semiconductor diode is zero.

Statement B: The current that exists under reverse-bias condition upto breakdown in *p-n* junction diode is called reverse saturation current.

(1) Both the statements A and B are correct

(2) Both the statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

CHEMISTRY

46. Given below are two statements.

Statement I: If an aqueous solution of 12.6 g oxalic acid dihydrate is made upto 1000 mL, then the volume of 0.1 N NaOH required to completely neutralize 10 mL of this oxalic acid dihydrate solution will be 20 mL.

Statement II: A gaseous mixture contains nitrogen gas and oxygen gas in the ratio of 1 : 2 by weight. Therefore, the ratio of moles of nitrogen gas to oxygen gas will be 4 : 7.

In the light of above statements, choose the **correct** answer from the option given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

47. During the Ostwald's process, if 34 g of NH_3 gas reacts with 44.8 L of $O_2(g)$ at STP, then consider the following statements.

$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$

(a) O_2 is limiting reagent

(b) 48.0 g of NO(g) is formed

(c) 143.36 L gaseous product present at the end of reaction at STP

(d) 53.76 L of H_2O (g) is formed at STP

The correct statements are

(1) (a) and (d) only

(2) (b), (c) and (d) only

(3) (a), (b) and (d) only

(4) (b) and (c) only

48. Given below are two statements

Assertion (A): Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.

Reason (R): Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff}).

In the light of above statements, choose the correct option.

(1) Both (A) and (R) true and (R) is the correct explanation of (A)

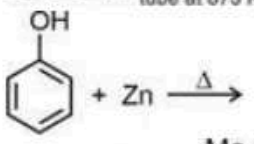
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

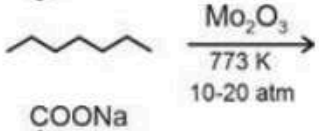
(3) (A) is true but (R) is false

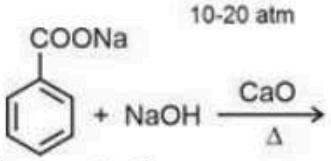
(4) Both (A) and (R) are false

49. Which of the following reactions will give benzene as the major product?

(a) $CH{\equiv}CH \xrightarrow[\text{tube at 873 K}]{\text{Red hot iron}}$

(b) 

(c) 

(d) 

Choose the correct option.

(1) (a), (b) and (d) only

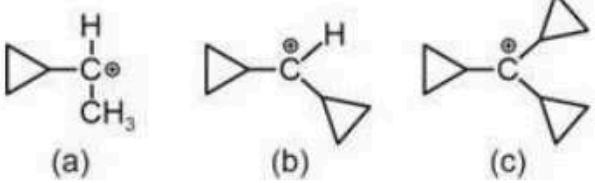
(2) (b) and (d) only

(3) (a), (b), (c) and (d)

(4) (a) and (b) only

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50. Arrange the following carbocations in their decreasing order of stability.



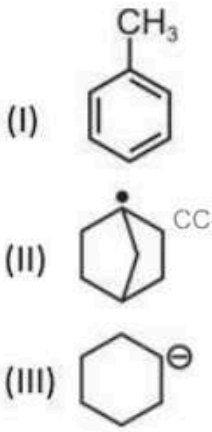
(1) (a) > (b) > (c)

(2) (c) > (a) > (b)

(3) (c) > (b) > (a)

(4) (b) > (a) > (c)

51. Consider the following compounds



Hyperconjugation occurs in

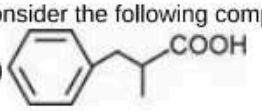
(1) (I) and (III) only

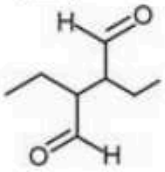
(2) (I) and (II) only

(3) (III) only

(4) (I) only

52. Consider the following compounds and their IUPAC names

(a)  2-Methyl-3-phenyl propanoic acid

(b)  Hexan-3,4-dial

(c) $HC \equiv C - CH_2 - CH_2 - CH = CH_2$ Hex-1-en-5-yne

The correct IUPAC names of the compounds are given as

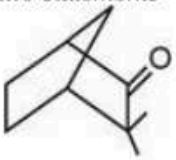
(1) (b) and (c) only

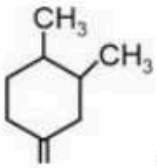
(2) (a) and (b) only

(3) (a), (b) and (c)

(4) (a) and (c) only

53. Given below are two statements

Statement I:  , compound can show tautomerism.

Statement II:  , compound does not show geometrical isomerism.

In the light of above statements, choose the correct answer from options given below.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

54. Considering the following statements

(a) PbS is yellow in colour

(b) On treating sodium fusion extract with sodium nitroprusside, appearance of a violet colour indicates the presence of sulphur in the compound

(c) AgBr is sparingly soluble in ammonium hydroxide.

The correct statements are

(1) (b) and (c) only

(2) (a) and (b) only

(3) (a), (b) and (c)

(4) (a) and (c) only

55. Assuming complete ionisation, same moles of which of the following compounds will require the highest amount of acidified $KMnO_4$ for complete oxidation in acidic medium?

(1) FeC_2O_4

(2) $Fe(NO_2)_2$

(3) $FeSO_4$

(4) $FeSO_3$

56. At 298 K, pH of a saturated solution of a diacidic base $X(OH)_2$ is 11. The solubility product (K_{sp}) of $X(OH)_2$ is

(1) 5.0×10^{-10}

(2) 7.0×10^{-9}

(3) 7.0×10^{-8}

(4) 4.0×10^{-9}

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57. A solution containing 1 mol CH_3COONa and 1 mol HCl per liter has pH 'a', while another solution containing 1 mol CH_3COONa and 1 mol CH_3COOH per liter has pH 'b'. Ratio of a : b will be (acid dissociation constant of CH_3COOH is K_a)

(1) 1 : 2

(2) pK_a : 2

(3) 2 : 1

(4) 2 : pK_a

58. The Born-Haber cycle for KCl is evaluated the following data:

$\Delta_fH°$for KCl = – 436.7 kJ mol^{-1}; $\Delta_{sub}H°$ for K = 89.2 kJ mol^{-1}; $\Delta_{ionization}H°$ for K = 419.0 kJ mol^{-1}; $\Delta_{bond}H°$ for Cl_2 = 243.0 kJ mol^{-1}; $\Delta_{lattice}H°(KCl)$ = – 717.8 kJ mol^{-1}

The magnitude of electron gain enthalpy of Cl(g) in (kJ mol^{-1}) will be

(1) 332.4

(2) 348.6

(3) 352.7

(4) 378.9

59. In the following reaction

$$\text{Toluene} \xrightarrow[273-283\ K]{\text{(i) } CrO_3, (CH_3CO)_2O,} \underset{(\text{Major})}{P}$$

(ii) H_3O^+, Δ

(iii) Acetophenone, $\overline{O}H$

(iv) Δ

Choose the incorrect statement about the above product (P).

(1) It contains ten degree of unsaturation

(2) It is $\alpha - \beta$ unsaturated carbonyl compound

(3) It undergoes Wolff-Kishner reduction, gives 1,3-Diphenylpropene

(4) It can be prepared by reaction of hex-4-enenitrile with diisobutylaluminium hydride followed by hydrolysis

60. Consider the following statements

(a) The correct order of bond length is $Li_2 > B_2 > F_2 > H_2$.

(b) Addition of two electrons in oxygen molecule results in increase of bond length.

(c) Conversion of NO $\rightarrow$ NO^+; electron is removed from π^* antibonding molecular orbital.

(d) The correct order of carbon-fluorine (C-F) bond length in CH_3F, CH_2F_2, CHF_3 and CF_4 molecule is $CH_3F > CH_2F_2 > CHF_3 > CF_4$.

(e) The correct order of O - N - O bond angles in the given species is $NO_2 > NO_2^+ > NO_2^-$.

The correct statements are

(1) (b), (c) and (e) only

(2) (a), (b) and (c) only

(3) (a), (b), (c) and (d) only

(4) (d) and (e) only

61. Match List-I with List-II

	List-I (Species)		List-II (Bond angle)
(a)	BF_3	(i)	Contains only 180° and 90° bond angle
(b)	XeF_4	(ii)	Contains only 120° bond angle
(c)	PCl_5	(iii)	Contains 90°, 180° and 120° bond angles
(d)	XeF_2	(iv)	Contains only 180° bond angle

The correct match is

(1) a(ii), b(iii), c(i), d(iv)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iv), d(iii)

62. $CoCl_3 \cdot xNH_3$ forms an octahedral complex (A). 0.2 molal aqueous solution of complex (A) shows a depression in freezing point of 1.116°C. (Assume 100% ionization of complex and K_f of water is 1.86 K $molal^{-1}$)

Select the correct answer from the options given below

(1) The molecular formula of the complex (A) is $[Co(NH_3)_6]Cl_3$

(2) If one NH_3 ligand of complex (A) is replaced by H_2O then it will show optical isomerism

(3) Magnetic moment of complex (A) is $\sqrt{35}$ BM

(4) If two NH_3 ligands of complex (A) are replaced by two water molecules then total three stereoisomers are possible

63. In which of the following options, the order of arrangement does not follow the variation of property indicated against it?

(1) Li < Na < K < Rb (increasing metallic radius)

(2) $Mg^{2+} < Na^+ < F^- < O^{2-}$ (increasing ionic size)

(3) Tl < Ga < Al < B (Increasing order of first ionisation energy)

(4) K < Na < Li < H (Increasing negative electron gain enthalpy)

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64. The plot of osmotic pressure (π) vs concentration (mol L^{-1}) for a solution (containing non-electrolytic solute) gives a straight line with slope 26.24 L bar mol^{-1}. The temperature at which osmotic pressure measurement is done is
(use universal gas constant R = 0.082 L bar mol^{-1} K^{-1})

(1) 47°C

(2) 320°C

(3) 2.15°C

(4) 21.5°C

65. Vapour pressure of a binary solution of two volatile liquids A and B is given by $P_T = (100 + 200\ X_A)$ mm of Hg. Vapour pressure of pure liquid A and B respectively are

(1) 300 mm of Hg and 100 mm of Hg

(2) 400 mm of Hg and 200 mm of Hg

(3) 100 mm of Hg and 300 mm of Hg

(4) 200 mm of Hg and 400 mm of Hg

66. Match electrolyte in list-I with amount of charge required to deposit 1 mol metal in list-II.

List-I	List-II
a. Aq. $CuSO_4$	(i) 2 F
b. Molten $AlCl_3$	(ii) 1 F
c. Molten AgBr	(iii) $\frac{8}{3}$ F
d. Motlen Fe_3O_4	(iv) 3 F

Correct match is

(1) a(i), b(iv), c(ii), d(iii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(i), b(iii), c(ii), d(iv)

67. If $E^{\circ}_{Fe^{3+}/Fe^{2+}} = x_1$ V and $E^{\circ}_{Fe^{3+}/Fe} = x_2$ V then $E^{\circ}_{Fe^{2+}/Fe}$ (in volt) will be

(1) $(x_1 + x_2)$

(2) $(x_1 - x_2)$

(3) $(3x_1 - 2x_2)/2$

(4) $(3x_2 - x_1)/2$

68. If 60% of a first order reaction was completed in 60 minutes, then 50% of same reaction would be completed in

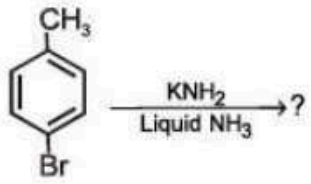
(1) 45.3 minute

(2) 62.2 minute

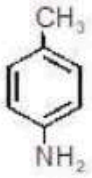
(3) 52.1 minute

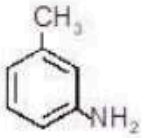
(4) 60 minute

69.

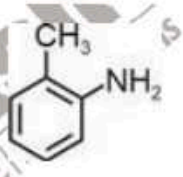


Major product is

(1) 

(2) 

(3) 

(4) 

70. The n-factor for Ferrocyanide in the given reaction is
$[Fe(CN)_6]^{4-} \rightarrow Fe^{3+} + CO_2 + NO_3^-$

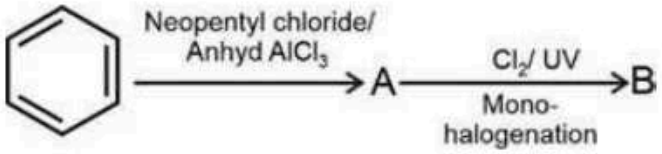
(1) 56

(2) 61

(3) 32

(4) 11

71.



Total number of isomers (including stereo isomers) of B obtained in the above reaction are

(1) 2

(2) 3

(3) 4

(4) 5

72. Given below are the two statements

Statement I: Amylopectin is insoluble in water and constitutes about 80-85% of starch

Statement II: Chemically amylopectin is a long unbranched chain with 200-1000 α-D-(+)-glucose units held together by C1-C4 glycosidic linkage.

In the light of above statements, choose the most appropriate answer from the options given below

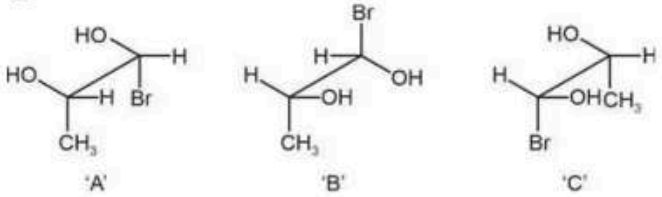
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

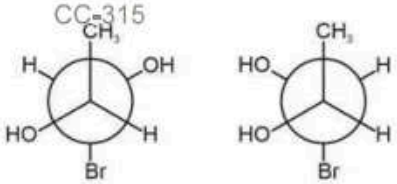
(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

73. Consider the given statements based on 'A', 'B', 'C', 'D' and 'E'



'A' 'B' 'C'



'D' 'E'

(a) 'A' and 'C' are enantiomers
(b) 'B' and 'C' are identical
(c) 'C' and 'D' are diastereomers
(d) 'C' and 'E' are enantiomers
The correct statement(s) is/are

(1) (a) and (b) only

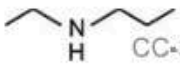
(2) (b) and (c) only

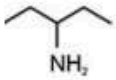
(3) (c) only

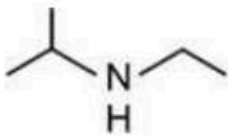
(4) (b), (c) and (d) only

74. A compound 'X' having molecular formula $C_5H_{13}N$ on reaction with Hinsberg's reagent gives compound 'Y' which dissolves in aqueous alkali. The structure of the compound 'X' could be

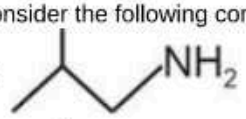
(1) 

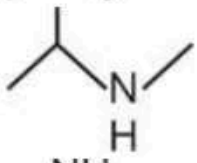
(2) 

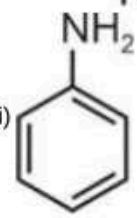
(3) 

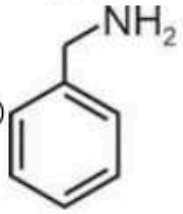
(4) 

75. Consider the following compounds

(i) 

(ii) 

(iii) 

(iv) 

The compound(s) which give positive isocyanide test is/are

(1) (ii) and (iii) only

(2) (i) only

(3) (i), (iii) and (iv) only

(4) (i) and (ii) only

76. Which of the following organic compound(s) undergo(es) disproportionation reaction with conc. NaOH?
(A) Propionaldehyde
(B) Formaldehyde
(C) Butanone
(D) Benzaldehyde

(1) (A) only

(2) (B) and (D) only

(3) (A), (B) and (C) only

(4) (A), (C) and (D) only

77. Given below are two statements:

Statement I: Methanal and ethanol can be distinguished by Iodoform test.

Statement II: Ethanol gives yellow precipitate of CHI_3 on Iodoform reaction.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement I and Statement II are false

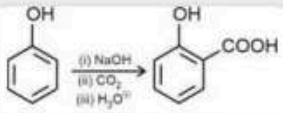
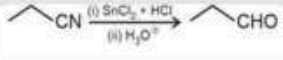
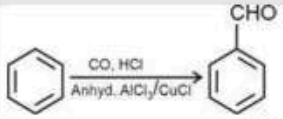
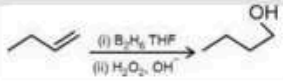
(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are true

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78. Match List-I with List-II

	List-I (Reaction)		List-II (Name of Reaction)
a.		(i)	Hydroboration oxidation reaction
b.		(ii)	Gatterman-Koch reaction
c.		(iii)	Stephen reaction
d.		(iv)	Kolbe's reaction

Choose the correct option.

(1) a(ii), b(iii), c(iv), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

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79. Given below are the two Statements

Statement I: Tert-Butyl alcohol on reaction with copper at 573 K gives 2-Methylpropene as major product.

Statement II: Propan- 1-ol on reaction with PCC gives propanoic acid as major product.

In light of above statements, choose the **correct** answer

(1) Statement I is Correct but Statement II is incorrect.

(2) Both Statement I and Statement II are Correct.

(3) Statement I is incorrect but Statement II is Correct.

(4) Both Statement I and Statement II are incorrect.

80. Consider the following statements

(a) SO_2 reacts with chlorine in the presence of charcoal to give thionyl chloride.

(b) SO_2 is used in refining petroleum and sugar.

(c) Moist sulphur dioxide behaves as reducing agent.

(d) SO_2 is a resonance hybrid of three canonical forms.

The correct statements are

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (b) and (c) only

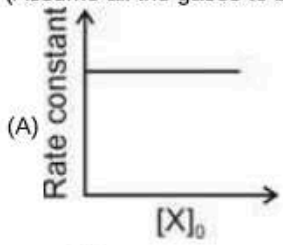
(4) (a) and (c) only

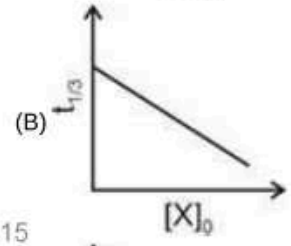
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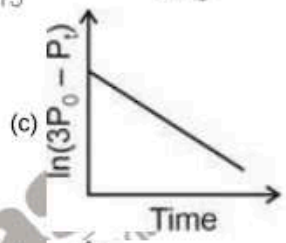
81. For a first order reaction,

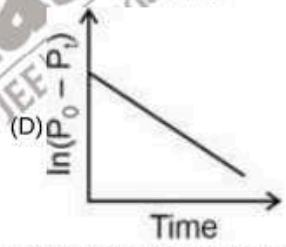
$X(g) \rightarrow 2Y(g) + Z(g)$ at constant volume and 300 K, the total pressure at the beginning and at time t are P_0 and Pt, respectively. Initially, only X is present with concentration $[X]_0$ and $t_{1/3}$ is the time required for partial pressure of X to reach $\frac{1}{3}$rd of it's initial value. The correct graph is/are

(Assume all the gases to be ideal gases)

(A) 

(B) 

(c) 

(D) 

Choose the correct option

(1) (A), (B) and (C) only

(2) (A) and (C) only

(3) (B) and (D) only

(4) (A) and (D) only

82. Consider the following statements regarding orthoboric acid.

(A) It can be prepared by acidifying an aqueous solution of borax.

(B) It is sparingly soluble in water but highly soluble in hot water.

(C) It behaves as a weak acid in water due to self-ionization.

(D) Acidity of an aqueous solution of orthoboric acid increases upon addition of cis-diol.

Choose the correct statements.

(1) (A) and (B) only

(2) (A), (B) and (C) only

(3) (C) and (D) only

(4) (A), (B) and (D) only

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83. The pair of lanthanoid ions which is diamagnetic in nature is

(1) Ce^{4+} and Lu^{3+}

(2) Sm^{3+} and Yb^{2+}

(3) Gd^{3+} and Eu^{3+}

(4) Pm^{3+} and Sm^{3+}

84. Correct order of C–O bond strength for the given species is

$\underset{(i)}{[Cr(CO)_6]}$ $\underset{(ii)}{[V(CO)_6]^-}$ $\underset{(iii)}{[Mn(CO)_6]^+}$

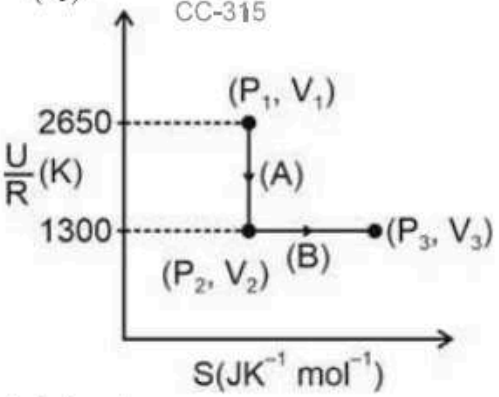
(1) (ii) > (iii) > (i)

(2) (ii) > (i) > (iii)

(3) (i) > (ii) > (iii)

(4) (iii) > (i) > (ii)

85. One mole of an ideal gas at 900 K undergoes two reversible processes 'A' followed by 'B', as shown below. If work done by the gas in two processes are same, then the value of $\ln\left(\frac{V_3}{V_2}\right)$ is



U - Internal energy
S - Entropy
P - Pressure
V - Volume
R - Gas constant

Molar heat capacity of gas at constant volume $C_{v,m} = \frac{5}{2}R$

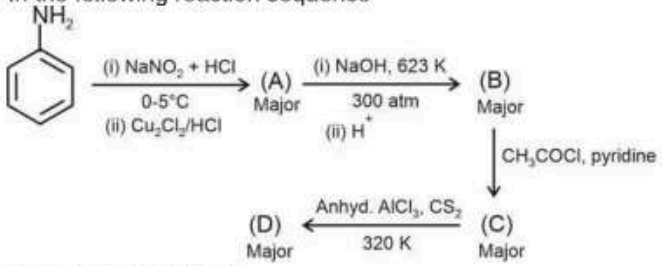
(1) 2

(2) 3.75

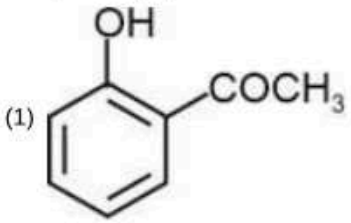
(3) 6

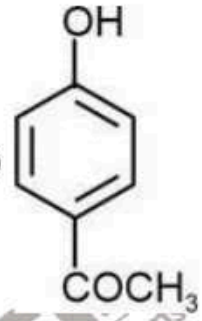
(4) 10

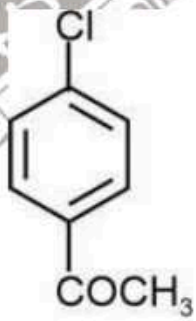
86. In the following reaction sequence



The product (D) will be

(1)


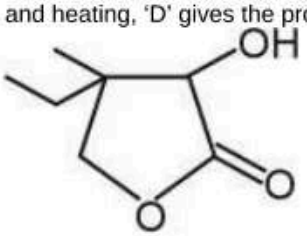
(2)


(3)


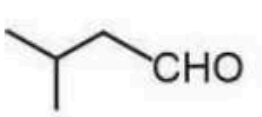
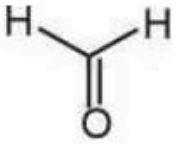
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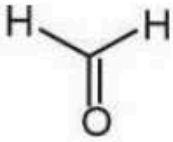

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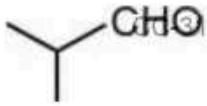
87. Two aliphatic aldehydes 'A' and 'B' react in the presence of aqueous K_2CO_3 to give compound 'C', which upon treatment with HCN provides compound 'D'. On acidification and heating, 'D' gives the product shown below:

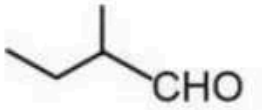
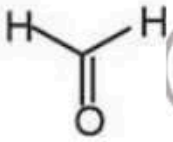


Compounds A and B respectivley are

(1)  and 

(2)  and 

(3)  and $CH_3 - CHO$

(4)  and 

88. Match electronic transition in List-I with total spectral lines observed for H atom sample in List-II.

	List-I		List-II
(a)	n = 3 to n = 1	(i)	6
(b)	n = 4 to n = 1	(ii)	10
(c)	n = 7 to n = 3	(iii)	3
(d)	n = 7 to n = 1	(iv)	21

The correct match is

(1) (a)(i), (b)(iii), (c)(ii), (d)(iv)

(2) (a)(ii), (b)(i), (c)(iii), (d)-(iv)

(3) (a)(iv), (b)(ii), (c)(i), (d)(iii)

(4) (a)(iii), (b)(i), (c)(ii), (d)(iv)

89. Match List-I with List-II.

List-I Cation group number	List-II Group reagent
a. Group-I	(i) H_2S gas in presence of dil.HCl
b. Group-II	(ii) H_2S in presence of NH_4OH
c. Group-III	(iii) Dilute HCl
d. Group-IV	(iv) NH_4OH in presence of NH_4Cl

Choose the correct answer from the options given below.

(1) a(iv), b(ii), c(iii), d(i)

(2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

90. Which of the following alcohols give blue colour in Victor Meyer's test?
(a) 3-Methylbutan-2-ol
(b) Cyclohexanol
(c) Isobutanol
(d) Neopentanol
(e) 2-Methylpentan-3-ol
The correct option is

(1) (b), (d) and (e) only

(2) (a), (b), (c) and (d) only

(3) (a), (b) and (e) only

(4) (c) and (d) only

BIOLOGY

91. A is the family of housefly, B is the class of wheat, C is the order of man and D is the genus of mango. Then find out the correct categories in the sequence of D, A, B and C respectively.

(1) Muscidae, Monocotyledonae, Primata, *Mangifera*

(2) *Mangifera*, Muscidae, Monocotyledonae, Primata

(3) Muscidae, Dicotyledonae, Insecta, Man

(4) Dicotyledonae, Primata, *Mangifera*, Mammalia

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92. Identify the following statements as True **(T)** or False **(F)** and select the **correct** option.
A. In lichens, the fungal component is known as mycobiont.
B. Bladderwort, Venus fly trap and *Cuscuta* are insectivorous plants.
C. In imperfect fungi, reproduction occurs only by sexual spores known as oospores.
D. Plant affected by viruses can have symptoms like leaf rolling and curling.

(1) A-(T), B-(F), C-(T), D-(T)

(2) A-(F), B-(T), C-(F), D-(F)

(3) A-(T), B-(T), C-(F), D-(T)

(4) A-(T), B-(F), C-(F), D-(T)

93. Observe the following diagram.



Select the **correct** statement for this organism.

(1) This organism can oxidise various inorganic substances and use the released energy for their ATP production.

(2) It is helpful in production of antibiotics such as penicillin.

(3) It reproduces sexually by a process involving cell fusion and zygote formation.

(4) This organism can fix both atmospheric nitrogen and carbon dioxide.

94. Sleeping sickness is caused by

(1) *Paramoecium*

(2) *Amoeba*

(3) *Entamoeba*

(4) *Trypanosoma*

95. Read the following statements w.r.t. gymnosperms.
(a) Leaves in gymnosperms have thick cuticle and sunken stomata.
(b) In *Cycas*, male or female reporductive bodies may be borne on the same tree.
(c) *Ginkgo biloba* have fan shaped leaves.
(d) Microsporophylls are compactly arranged on central axis to form a microsporangiate strobili.
(e) Two kinds of spores are produced within sporangia that are borne on male strobili.
Select the option with **correct** set of statements.

(1) (a), (c), (d), & (e) only

(2) (a), (c) & (d) only

(3) (a), (b) & (c) only

(4) (c), (d), & (e) only

96. How many of the given plants produce seeds and also have archegonia?
a. *Pinus*
b. *Funaria*
c. *Selaginella*
d. Mustard
e. *Porphyra*
f. *Cycas*

(1) Two

(2) Three

(3) Four

(4) Five

97. Chemotaxonomy is based on

(1) Chemical constituents of a plant

(2) Chromosome number and structure

(3) Codes and numbers assigned to plants

(4) Phylogenetic system of classification

98. Match **List-I** with **List-II**

List-I		**List-II**
a. Argemone	(i)	Ovules are attached to the placenta present in axial position in a multilocular ovary
b. Marigold	(ii)	The placenta develops at the base of ovary
c. Dianthus	(iii)	Ovules are borne on central axis and septa are absent in ovary
d. Lemon	(iv)	Ovary become two-chambered due to formation of false septum

Choose the **correct** option

(1) a(i), b(iii), c(ii), d(iv)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iii), b(i), c(ii), d(iv)

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99. A compound leaf differs from a simple leaf in

(1) Bearing axillary buds in axil of its leaflets

(2) Having reticulate venation

(3) Having a number of leaflets as incisions of the lamina reach upto the midrib

(4) Bearing a bud in the axil of the petiole

100. Non-endospermous seed are found in the plants like

(1) Bean and gram

(2) Bean and coconut

(3) Maize and barley

(4) Onion and maize

101. Select the **correctly** matched pair.

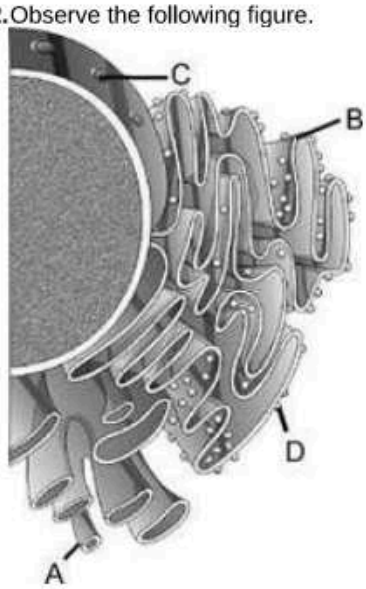
(1) Epiblema – Forms unicellular elongations that helps to absorb water and minerals from the soil.

(2) Cuticle – Water impermeable substance that forms a thick layer to cover the root, stem and leaves of higher plants.

(3) Palisade parenchyma – Constituent cells are placed beneath the abaxial epidermis and are arranged vertically and parallel to each other.

(4) Guard cell – Specialised cell that surrounds the non-photosynthetic subsidiary cells.

102. Observe the following figure.



Select the option that **correctly** depicts the labels given in the figure.

(1) Label A – It is extensive and continuous with the outer membrane of the nucleus.

(2) Label B – It synthesises lipid-like steroidal hormones in animal cells

(3) Label C – It provides the passage through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm.

(4) Label D – It is a glycolipid.

103. Pericycle forms semi-lunar patches of sclerenchyma cells in

(1) Dicot roots

(2) Monocot stems

(3) Dicot stems

(4) Monocot roots

104. For the first time, Rudolf Virchow told that *Omnis cellula-e cellula*. It means

(1) All living organisms are composed of cells

(2) Each cell has organelles that function in coordinated manner

(3) All cells arise from pre-existing cells

(4) Activities of an organism are the result of activities of its constituent cells

105. Who studied different type of animal cells and reported that cell had a thin outer layer?

(1) Robert Hooke

(2) Robert Brown

(3) Matthias Schleiden

(4) Theodore Schwann

106. The innermost layer of walls of blood vessels is composed of

(1) Ciliated columnar epithelium

(2) Columnar epithelium

(3) Cuboidal epithelium

(4) Squamous epithelium

107. In the following question, a statement of assertion (A) is followed by a statement of reason (R).

Assertion (A): CO_2 can diffuse through the diffusion membrane more rapidly than O_2 per unit difference in partial pressure.

Reason (R): Solubility of CO_2 in plasma is 20-25 times higher than that of O_2.

In the light of above statements, select the **correct** option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

108. Which of the following is a primary metabolite?

(1) Carotenoids

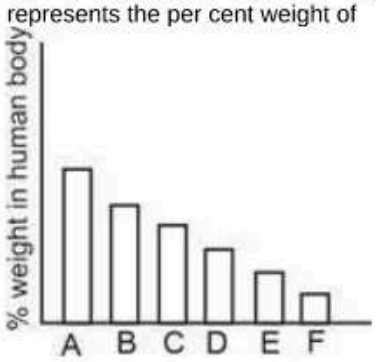
(2) Anthocyanins

(3) Amino acids

(4) Rubber

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109. Percentage weight of different elements present in the human body is shown below with the help of a bar graph. If bar 'B' represents per cent weight of carbon, then 'A' represents the per cent weight of



(1) Silicon

(2) Nitrogen

(3) Magnesium

(4) Oxygen

110. Choose the **odd** one w.r.t. groups present in serine that are responsible for its chemical and physical properties.

(1) Amino

(2) Carboxyl

(3) Hydroxy methyl

(4) Sulfur

111. Which option best describes the tertiary level of protein structure?

(1) The association of multiple polypeptide chains through weak interactions.

(2) The specific three-dimensional conformation of a fully folded polypeptide, as exhibited by enzymes.

(3) The arrangement of a polypeptide chain into left-handed helices, just like an alpha helix or beta-pleated sheet.

(4) The overall shape formed when two or more polypeptide subunits come together, as seen in haemoglobin.

112. Total number of carbon atoms present in arachidonic acid and palmitic acid respectively are

(1) 16 and 20

(2) 18 and 20

(3) 20 and 16

(4) 22 and 16

113. All mammals, without any exception, are characterised by

(1) Viviparity and biconcave red blood cells

(2) Heterodont teeth and 12 pairs of cranial nerves

(3) Extra-abdominal testes and a four-chambered heart

(4) A muscular diaphragm and mammary glands

114. Read the following statements A and B. Choose the **correct** option.

Statement A : Members of the class chondrichthyes show internal fertilization and many of them are viviparous.

Statement B : Due to the absence of air bladder, cartilaginous fishes have to swim constantly to avoid sinking.

(1) Statement A is correct but B is incorrect.

(2) Statement A is incorrect but B is correct.

(3) Both statements A and B are correct.

(4) Both statements A and B are incorrect.

115. Common feature of *Ascidia, Amphioxus*, Myxine, *Rana, Aptenodytes* and Blue whale is

(1) Presence of cloaca

(2) Presence of paired and unpaired appendages

(3) Presence of lateral line system

(4) Presence of notochord ventral to nerve cord during their embryonic development

116. The smallest living cell

(1) Is a prokaryote

(2) Has mitochondria but no chloroplast

(3) Lacks cell membrane

(4) Lacks genetic material

117. Metaphase I is different from metaphase II as in the former

(1) Bivalent chromosomes align on the equatorial plate

(2) Homologous chromosomes separate from each other

(3) Crossing over between non-sister chromatids of homologous chromosomes occur

(4) Metaphasic plate is not formed

118. Given are two statements Assertion (A) and Reason (R).

Assertion (A): DNA replication does not take place just before meiosis II.

Reason (R): DNA is already present in double amount in each chromosome just before meiosis II as compared to that of G_1 phase.

In the light of above statements select the **correct** option.

(1) Both (A) and (R) are false

(2) Both (A) and (R) are true and (R) is correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

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119. Observe the following events of M phase of cell cycle.
a. Spindle fibres attach to kinetochores of chromosomes.
b. Centrosome begins to move towards opposite poles of the cell.
c. Chromosomes begin their migration towards the two opposite poles.
d. Appearance of a furrow in the plasma membrane.
e. Chromosome cluster at opposite spindle poles and their identity is lost as discrete elements.
Select the option that depicts the **correct** sequence of the occurrence of these events.

(1) a → d → b → e → c

(2) b → e → c → a → d

(3) b → a → c → e → d

(4) c → a → d → b → e

120. Select the **correct** option w.r.t. products of light reaction of photosynthesis.

(1) Glucose and oxygen

(2) ATP, NADH, CO_2 and H_2O

(3) ATP, NADPH and O_2

(4) Glucose, ATP and O_2

121. Plants like *Sorghum*

(1) Have Kranz anatomy in their leaves

(2) Show oxygenase activity of RuBisCO

(3) Lack chloroplasts in bundle sheath cells

(4) Lack photochemical phase of photosynthesis

122. During aerobic respiration which of the following conversions shows substrate level phosphorylation?

(1)	Oxalosuccinic acid	→	a-ketoglutaric acid
(2)	Fumaric acid	→	Malic acid
(3)	Succinyl CoA	→	Succinic acid
(4)	Citric acid	→	Cis-Aconitic acid

(1) (1)

(2) (2)

(3) (3)

(4) (4)

123. If the amount of carbon dioxide evolved during respiration is more than the amount of oxygen consumed, then the respiratory substrate is

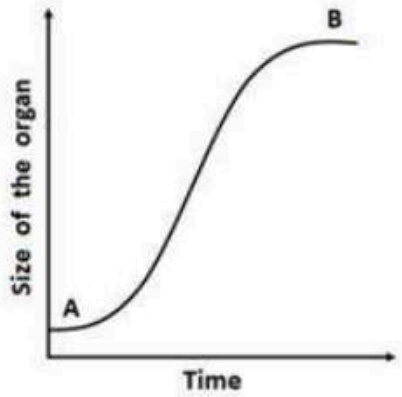
(1) Carbohydrate

(2) Tripalmitin

(3) Proteins

(4) Organic acids

124. Observe the graphical representation given below and select the correct option.



(1) The graph represents arithmetic growth of a plant organ.

(2) This growth pattern can be correctly represented by the equation, $L_t = L_0 + rt$

(3) The curve between the phases A and B represents the growth of a tree during unfavourable season.

(4) Label A represents lag phase of growth

125. Which of the following is **not true** w.r.t effects of auxins?

(1) Apical dominance

(2) Internode elongation in beet just prior to flowering

(3) Inhibition of growth of lateral buds

(4) Induction of flowering in pineapple

126. Read the given features and choose the **correct** option.
(a) Induction of flowering by a gaseous hormone.
(b) Polyembryony occurs due to formation of multiple embryo sacs arises from nucellar and integument cells.
(c) Presence of one megasporangium in an ovary.
(d) Fruit develops from monocarpellary superior ovary.
(e) Thalamus also contributes to fruit formation.
(f) Male gametophyte maintains viability for months as it belongs to family Solanaceae.
How many of the above statement(s) is/are **true** for *Mangifera indica*?

(1) One

(2) Four

(3) Two

(4) Three

127. Select the **incorrect** match from the following.

(1) Coleoptile – Encloses leaf primordia

(2) Epiblast – Remains of second cotyledon

(3) Scutellum – Cotyledon in monocots

(4) Coleorhiza – Portion of embryonal axis above the cotyledon

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128. Primary endosperm nucleus is produced by fusion of

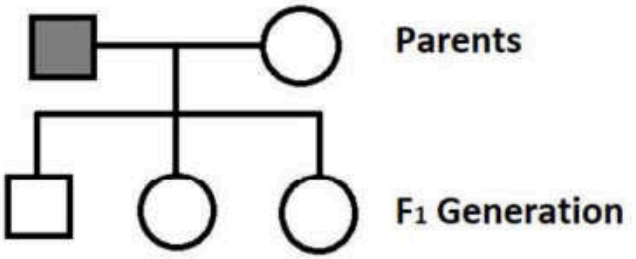
(1) Male gamete and egg

(2) Male gamete and one polar nucleus

(3) Two polar nuclei

(4) Male gamete and two fused polar nuclei

129. Consider the given pedigree chart.



If the male parent is colorblind as well as suffering from phenylketonuria and the female in F_1 generation got married to a male suffering from phenylketonuria but have normal vision, what would be true for F_2, generation?

(1) 50% of the male children will suffer with color blindness and 50% with phenylketonuria.

(2) All girls will be unable to discriminate between red and green colour.

(3) Chances of female progeny being colorblind is more than male progeny

(4) All the individuals which will be suffering from phenylketonuria will also be colourblind.

130. Four genes a, b, c and d are present on a chrosome. Percentage of recombination between a and b = 20%; a and d = 2%, c and b = 8% and d and b = 18%. Which of the following will be the sequence of genes on chromosome?

(1) adcb

(2) cdab

(3) dcab

(4) acbd

131. **Assertion (A):** In cockroach, each abdominal segment is covered by four hardened plates called sclerites.
Reason (R): Sclerites are joined to each other by a thin and flexible arthrodial membrane.
In the light of above statements, select the **correct** option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

132. Which of the following statements is/are **correct** about frog?

(a) Lymph is different from blood as it does not contain any protein
(b) RBCs in frog are enucleated to accommodate large amount of haemoglobin
(c) Sinus venosus receives blood through vena cava and pours it into right atrium
Select the **correct** option.

(1) (a) and (b)

(2) (b) and (c)

(3) (a) and (c)

(4) (c) only

133. The type of epithelium present in the inner lining of ducts of salivary glands is also found in the

(1) Blood vessels

(2) Pharynx

(3) Small intestine

(4) Loop of Henle

134. When oxygen-haemoglobin dissociation curve gets shifted towards the right side, it reflects all of the following, **except**

(1) Decreased affinity of haemoglobin for oxygen

(2) Dissociation of oxygen from haemoglobin

(3) Decrease in P_{50} value

(4) Less percentage saturation of haemoglobin with O_2

135. **Assertion (A):** In humans, left ventricle has the thickest wall in comparison with any other chamber of the heart.
Reason (R): It pumps oxygenated blood to the whole body via aorta.
In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) (A) is false but (R) is true

136. Erythroblastosis foetalis can be prevented by administering anti-Rh antibodies to

(1) Rh –ve mother post partum Rh +ve baby

(2) Rh +ve mother post partum Rh +ve baby

(3) Rh –ve mother post partum Rh –ve baby

(4) Rh +ve mother post partum Rh –ve baby

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137. Select the **mismatch** w.r.t. a cardiac cycle.

(1) Stroke volume : Approximately 70 mL blood

(2) Cardiac output : Averages 5000 mL or 5 litres in a healthy individual

(3) Pacemaker : Present at the top right corner of left atria

(4) Isovolumetric contraction : Atrio-ventricular and semilunar valves are closed

138. In human excretory system, all of the following are paired structures, **except**

(1) Kidney

(2) Renal pelvis

(3) Urethra

(4) Ureter

139. When you feel thirsty or there is dehydration, it results in increase in osmolarity of the blood above a physiological set point. And, this leads to the osmoreceptors of the hypothalamus to stimulate the pituitary to release

(1) Antidiuretic hormone (ADH)

(2) Atrial natriuretic factor

(3) Aldosterone

(4) Renin

140. The function of troponin and tropomyosin in skeletal muscles includes

(1) Sliding of myosin on actin for shortening of sarcomere

(2) Release of calcium after initiation of contraction

(3) Binding to myosin during contraction

(4) Masking of the myosin binding site present on actin when at rest, as they act as regulatory proteins

141. In *Paramoecium*, cilia help in

(1) Movement of tentacles

(2) Streaming of protoplasm

(3) Movement of food through cytopharynx

(4) Capturing of prey but not in locomotion

142. Comprehend the given statements:

(a) The neural system of all animals is composed of highly specialised cells called neurons.

(b) In *Hydra,* a network of interconnected nerve cells is present but it lacks a centralised brain and specialised sense organs.

(c) When an excitatory neurotransmitter excites the motor neuron, the neuronal membrane becomes highly permeable to certain cations.

Select the **correct** option.

(1) (a), (b) and (c) are correct

(2) Only (a) and (b) are correct

(3) Only (c) is correct

(4) (b) and (c) are incorrect

143. Choose the **correct** match w.r.t humans.

(1) Trophoectoderm - Inner layer of morula

(2) Chorionic villi – Finger-like projections at the end of fallopian tubes

(3) Colostrum - Enriched with IgA

(4) Implantation - Occurs only after one month of fertilization

144. During the later phase of pregnancy, relaxin is secreted by _____ and it plays a role in parturition by relaxing the ligaments and softening of the cervix in mother.

Select the **correct** option to fill in the blank.

(1) Hypothalamus

(2) Adenohypophysis

(3) Pars nervosa

(4) Ovary

145. A network of intra-testicular accessory ducts, named 'X', receives sperms directly from seminiferous tubules and structurally leads into ducts that leave the testis to open into a structure located on posterior surface of each testis. This 'X' is

(1) Vasa efferentia

(2) Rete testis

(3) Ejaculatory duct

(4) Epididymis

146. Which of the following is reusable contraceptive made of rubber and gives privacy to user?

(1) Condoms

(2) Lippes loop

(3) Cervical caps

(4) CuT

147. How many diseases given in the box below are considered as venereal diseases?

Gonorrohoea, Measles, Hepatitis-B, Diphtheria, Syphilis, Genital herpes, Genital warts, Dengue

Select the **correct** option.

(1) Four

(2) Five

(3) Seven

(4) Three

148. Big-Bang theory attempts to explain the

(1) Origin of life

(2) Origin of species

(3) Origin of Earth

(4) Origin of Universe

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149. In 1938, a fish caught in South Africa happened to be a coelacanth which was thought to be extinct. These animal called lobe fins evolved into first

(1) Vertebrates

(2) Gnathostomes

(3) Amphibians

(4) Chordates

150. Match the column I with column II.

Column I	Column II
a. *Homo erectus*	(i) More ape-like
b. *Homo habilis*	(ii) Average cranial capacity 1400cc
c. *Dryopithecus*	(iii) Probably ate meat
	(iv) Probably did not eat meat

Choose the correct option.

(1) a(iv), b(iii), c(i)

(2) a(iii), b(iv), c(i)

(3) a(ii), b(iv), c(i)

(4) a(iv), b(ii), c(i)

151. A physiological barrier to the entry of micro-organisms in human body is

(1) Skin

(2) Epithelium of GIT

(3) Tears

(4) Monocytes

152. Complete the analogy and select the **correct** option.
Ringworms : *Microsporum* : : Pneumonia : _____

(1) *Haemophilus influenzae*

(2) Rhinovirus

(3) *Salmonella typhi*

(4) *Trichophyton*

153. Which of the following enzymes exist in many bacteria where they function as a part of their defense mechanism, called the restriction-modification system?

(1) Restriction enzymes and DNA ligases

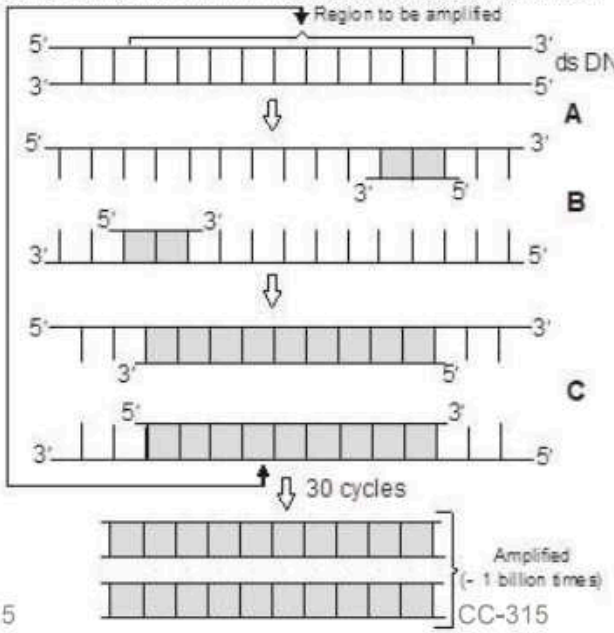
(2) Restriction enzymes and DNA polymerase

(3) Restriction enzymes and methylases

(4) Restriction enzymes and alkaline phosphatase

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154. Following diagram represents polymerase chain reaction with steps labelled as A, B and C. Choose the option representing the **correct** description of steps mentioned.



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(1) A - Denaturation – Occurs at 40°C

(2) B - Annealing – Requires activity of *Taq* polymerase

(3) C - Extension – Requires ribonucleotides

(4) A - Denaturation – Requires temperature above 90°C.

155. The letter 'R' in the name of restriction endonuclease *Eco* RI denotes

(1) Genus of the source organism

(2) Species of the source organism

(3) Strain of the source organism

(4) Order of discovery of the enzyme from the source organism

156. A dihybrid cross is performed between two pea plants with genotype BBYY and bbyy. The gene B controls synthesis of starch grains and gene Y controls seed colour. Choose the **correct** option w.r.t. the F_2 generation if F_1 progeny obtained by this cross is selfed.

(1) 25% of the progeny will be having wrinkled yellow seeds

(2) 12.5% of the seeds would be round-green with intermediate size starch grains.

(3) The phenotypic ratio obtained will be 3 : 1 : 4 : 2 :1

(4) 35% of the progeny of F_2 generation will be having more than two dominant alleles

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157. Human females have

(1) Autosomes + XX

(2) Autosomes + XY

(3) Autosomes + XO

(4) Autosomes + ZW

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158. State **True(T)** or **False(F)** to the given statements.

a. Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide.

b. Lambda phage has single-stranded linear DNA with 48502 nucleotides.

c. Rho factor (ρ) is required for termination of transcription

d. UTRs are present at both 5'-end after start codon and 3'-end before stop codon.

e. M. Wilkins and Rosalind Franklin had proposed a definitive double helix model of DNA on the basis of X-ray diffraction pictures

Choose the **correct** option.

	a	b	c	d	e
(1)	F	T	T	T	T
(2)	T	T	F	F	T
(3)	T	F	T	F	F
(4)	T	F	F	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

CC-315

159. In a 51 nm long *E. coli* genomic segment, adenine is 20% of the total nitrogenous bases. Calculate the total number of hydrogen bonds present in this segment. Choose the correct answer from the options given below.

(1) 150 hydrogen bonds

(2) 270 hydrogen bonds

(3) 300 hydrogen bonds

(4) 390 hydrogen bonds

160. An isolated root tip cell from *Vicia faba* plant is allowed to multiply for several generations in medium containing radiolabelled thymidine residues. A daughter cell (having 100% radiolabelled thymidine residues) is transferred to non-radioactive medium. If this cell completes 3 mitotic generations, in what per cent of daughter cells no radioactivity is detected?

(1) 25%

(2) 50%

(3) 75%

(4) Zero

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161. A transcription unit is 7000 nucleotides long. This unit may use __A__ nucleotides to make a polypeptide that can have __B__ amino acids.

Select the **correct** option to fill in the blanks A and B.

	A	B
(1)	3500	1750
(2)	7000	2330
(3)	2100	650
(4)	1400	500

(1) (1)

(2) (2)

(3) (3)

(4) (4)

162. In *lac* operon, permease is synthesized by gene

(1) *lac* z

(2) *lac* y

(3) *lac* a

(4) *lac* i

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163. Identify the **correct** match from the following.

(1)	Beetle with red and black markings	–	Prevents mosquitoes
(2)	*Bacillus thuringiensis*	–	Kills butterfly caterpillars
(3)	*Trichoderma*	–	Free-living bacteria
(4)	Baculovirus	–	Broad spectrum insecticidal virus

(1) (1)

(2) (2)

(3) (3)

(4) (4)

164. Complete the following table by selecting correct option for P, Q and R

Microbe	Product	Usage
Aspergillus niger	P	Flavouring and Preservation of food
Q	Acetic acid	R

(1) P – Citric acid R – Preparation of vinegar

(2) P – Citric acid Q – *Lactobacillus*

(3) R – Making curd Q – *Clostridium*

(4) R – Preparation of vinegar Q – *Lactobacillus*

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165. The intrinsic rates of natural increase for three populations; (A) Norway rate, (B) flour beetle and (C) human are 0.015, 0.12 and 0.0205 respectively. All three populations are assumed to be growing in the same habitat. If the value of N for all is exactly equal to K/2, which of the following depicts the correct order of these populations reaching the asymptote?

	First	Second	Last
(1)	A	B	C
(2)	C	B	A
(3)	B	A	C
(4)	B	C	A

(1) (1)

(2) (2)

(3) (3)

(4) (4)

166. In how many of the given interactions, at least one species is harmed?

Mutualism, Competition, Predation, Parasitism, Commensalism, Amensalism, Protocooperation

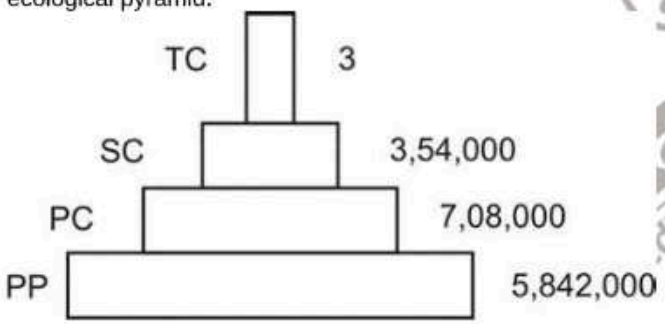
(1) Three

(2) Four

(3) Five

(4) Two

167. Choose the **incorrect** statement w.r.t. the following ecological pyramid.



(1) It represents the pyramid of number of a grassland and pond ecosystem

(2) PC can be occupied by grasshopper in a grassland ecosystem

(3) It represents the pyramid of number in a tree ecosystem

(4) TC are top carnivores in above given ecosystem

168. Which one is the major conduit of energy flow in aquatic ecosystem?

(1) Grazing food chain

(2) Detritus food chain

(3) Parasitic food chain

(4) Auxiliary food chain

169. "Biodiversity hotspots", are/show

(1) High degree of endemism

(2) Slow rate of extinction

(3) Less species richness

(4) Only five in India

170. Read the following Assertion (A) and Reason (R) and chose the **correct** option

Assertion (A) : Biologists are not sure about how many prokaryotic species there might be.

Reason (R) : Conventional taxonomic methods are not suitable for identifying the species of prokaryotes in nature and many of the species of prokaryotes are non culturable under laboratory conditions.

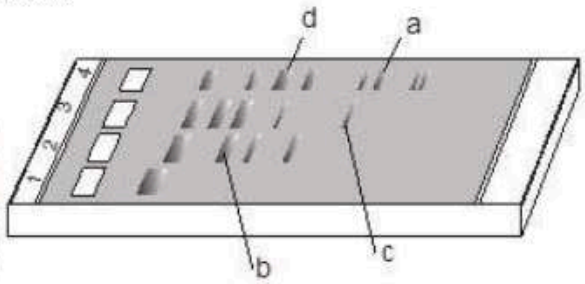
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

171. A typical agarose gel electrophoresis showing migration of digested DNA fragments is represented by the image given below:



Arrange the obtained fragments a, b, c and d in increasing order of their size.

(1) a, c, d, b

(2) a, d, c, b

(3) d, a, b, c

(4) a, c, b, d

172. Select the DNA sequence which can act as a recognition site of a restriction endonuclease

(1) 5′AACCGG3′
3′TTGGCC5′

(2) 5′GGTTGG3′
3′CCAACC5′

(3) 5′AAGGCT3′
3′TTCCGA5′

(4) 5′CTGCAG3′
3′GACGTC5′

173. Which of the following is an **incorrect** statement?

(1) Developing countries are rich in traditional knowledge and bio-resources

(2) There are an estimated 2,00,000 varieties of rice in India

(3) For chemical safety testing, transgenic animals are made less sensitive to toxic substances

(4) The Indian parliament has recently cleared the second amendment of the Indian Patents Bill

174. A structural syncytium is found in

(1) Cardiac muscles

(2) Muscles of small intestine

(3) Muscles of blood vessels

(4) Skeletal muscles

175. Which of the following is **not** a polymer of glucose?

(1) Starch

(2) Inulin

(3) Glycogen

(4) Cellulose

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176. Which of the following is an example of artificial selection?

(1) Industrial melanism

(2) Development of antibiotic resistance in bacteria

(3) Generation of different breeds of dogs

(4) Sickle cell anaemia in African population

177. The hormone which acts on both pancreas and gall bladder and stimulates the secretion of pancreatic enzymes and bile juice respectively is

(1) CCK

(2) GIP

(3) Gastrin

(4) Secretin

178. The mechanism by which 'X', secretion from α-cells of Islets of Langerhans, lead to an increase in the plasma glucose concentration involves

(1) Its binding to DNA in the nucleus of target cells

(2) Its binding to receptors present in cytoplasm of target cells

(3) Glycogenolysis in liver cells

(4) Activation of G-protein to produce hormone in target cells by interacting *via* intracellular receptors.

179. A female patient 'X' goes to the clinic of an endocrinologist due to the swelling of the butterfly-shaped gland present in her neck region. Her blood test results show low T_4 as well as TSH levels.
Later, TSH is injected in her which leads to increase in the levels of T_4. Based on the given scenario, which endocrine gland/part was defective in 'X'?

(1) Pars distalis

(2) Thyroid gland

(3) Adrenal gland

(4) Pars nervosa

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180. Which division of the nervous system would likely be activated, if a student enters the examination hall to sit for his science paper provided to him and suddenly realises that he had mistakenly prepared for mathematics exam instead?

(1) Somatic neural system

(2) Parasympathetic neural system

(3) Sympathetic neural system

(4) Only CNS present below the medulla oblongata

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